

Deterministic Optimization

Robust Optimization

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Robust Linear Program

Robust Optimization

Learning Objectives for this lesson

- Formulate general robust linear constraints
- Explore different types of uncertainty sets
- Discover a way to reformulate robust linear constraints as deterministic linear constraints

Example: Drug Production

$$\min \quad 100 \cdot \text{RawI} + 199.9 \cdot \text{RawII} + 700 \cdot \text{DrugI} + 800 \cdot \text{DrugII} - (6200 \cdot \text{DrugI} + 6900 \cdot \text{DrugII})$$

$$\text{s.t.} \quad \text{ActiveI} \cdot \text{RawI} + \text{ActiveII} \cdot \text{RawII} - 0.5 \cdot \text{DrugI} - 0.6 \cdot \text{DrugII} \geq 0$$

$$\forall \text{ActiveI} \in [0.00995, 0.01005], \text{ActiveII} \in [0.0196, 0.0204]$$

$$\text{RawI} + \text{RawII} \leq 1000$$

$$90 \cdot \text{DrugI} + 100 \cdot \text{DrugII} \leq 2000$$

$$40 \cdot \text{DrugI} + 50 \cdot \text{DrugII} \leq 800$$

$$100 \cdot \text{RawI} + 199.9 \cdot \text{RawII} + 700 \cdot \text{DrugI} + 800 \cdot \text{DrugII} \leq 100000$$

$$\text{RawI}, \text{RawII}, \text{DrugI}, \text{DrugII} \geq 0.$$

Robust Linear Constraint

- Deterministic linear constraint:
 - $a_0^\top x \geq b$
 - Solution x satisfies the linear inequality for one parameter value a_0
- Parameter a may be uncertain and lives in a set $a \in U$
 - In drug production, a is content of active agent and U is an interval
 - U is called the **uncertainty set**
- Robust linear constraint
 - $a^\top x \geq b \quad \forall a \in U$
 - Robust feasibility: Solution x satisfies linear inequality for all possible values of a in the uncertainty set U

Reformulation of Robust Constraint

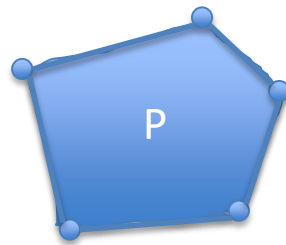
- Equivalent reformulation:
 - $a^\top x \geq b \quad \forall a \in U$ is equivalent to
 - $\min_{a \in U} a^\top x \geq b$
 - The minimization on the left is over variable a , where x is treated as fixed
- If U contains an infinite number of elements, robust constraint $a^\top x \geq b \quad \forall a \in U$ becomes a conjunction of infinite number of constraints, each for a value of a in U
- One robust linear constraint corresponds to an infinite number of deterministic linear constraints

Finite Uncertainty Set

- If U is a set of finite elements, i.e. a finite uncertainty set
 - $U = \{a_1, a_2, \dots, a_N\}$
- Robust constraint $a^\top x \geq b \quad \forall a \in U$ becomes
 - $$\begin{cases} a_1^\top x \geq b \\ a_2^\top x \geq b \\ \dots \\ a_N^\top x \geq b \end{cases}$$

Convex Hull of Scenarios

- Suppose an uncertainty set $U := \text{conv}\{a_1, \dots, a_N\}$
- Robust constraint $\min_{a \in U} a^\top x \geq b$ becomes
- $\min_{a \in \text{conv}\{a_1, \dots, a_N\}} a^\top x \geq b$
- Which is equivalent to
- $\min_{a \in \{a_1, \dots, a_N\}} a^\top x \geq b$
- Why?
- Because LP optimal solution is an extreme point



Polyhedral Uncertainty Sets

- Suppose an uncertainty set U is a polyhedron and is given by the halfspace representation: $U = \{a : Ba \leq g\}$
- Robust constraint becomes

- $\min_{a:Ba \leq g} a^\top x \geq b$

- How should we further reformulate this?

- The answer is linear programming duality!

- $\min_{a:Ba \leq g} a^\top x = \max_{y: B^\top y = x, y \leq 0} g^\top y \geq b$

- The final step: $\begin{cases} g^\top y \geq b \\ B^\top y = x \\ y \leq 0 \end{cases}$  A finite number of linear constraints!

Drug Production Example

$$\begin{aligned} &ActiveI \cdot RawI + ActiveII \cdot RawII - 0.5 \cdot DrugI - 0.6 \cdot DrugII \geq 0 \\ &ActiveI \in [0.00995, 0.01005], ActiveII \in [0.0196, 0.0204] \end{aligned}$$



$$\begin{aligned} &\min ActiveI \cdot RawI + ActiveII \cdot RawII - 0.5 \cdot DrugI - 0.6 \cdot DrugII \geq 0 \\ &s.t. ActiveI \in [0.00995, 0.01005], ActiveII \in [0.0196, 0.0204] \end{aligned}$$



$$0.00995 \cdot RawI + 0.0196 \cdot RawII - 0.5 \cdot DrugI - 0.6 \cdot DrugII \geq 0$$

Summary

- We learned:
 - The concept of uncertainty sets and different types of uncertainty sets
 - Reformulation of robust constraints